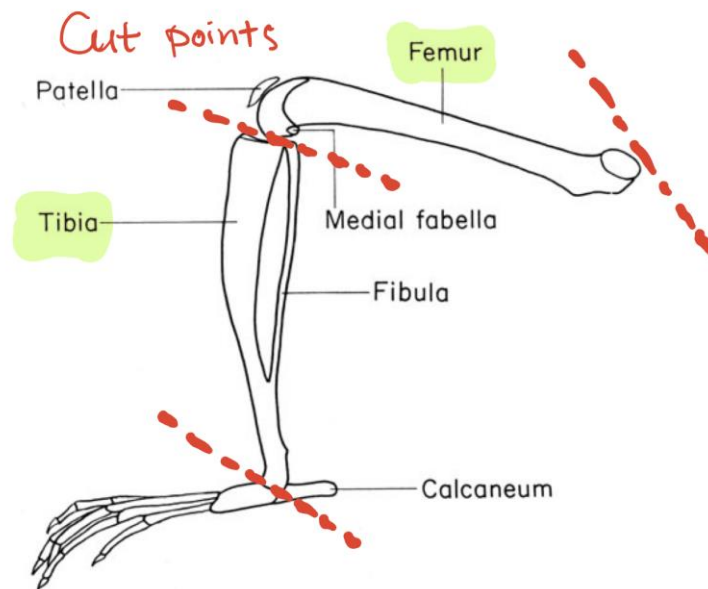


Hsu Lab bone-marrow derived macrophages (BMDM) generation

Based on protocols from Leitinger lab

1. On day 0, extract bone marrow cells from the femurs and tibias of mice
 - Remove skin, cut away both hind legs at the hip joint
 - Soak legs in 70% EtOH for 1 minute, then wash with PBS
 - Using scissors, cut away muscles and isolate femurs and tibia
 - Cut off each end of the bone and **flush with serum free RPMI** through the 100 μ m mesh with 26½ gauge needle. (flushing out is more gentle than centrifuging bone marrow cells out)
 - Transfer the bone marrow content to 50 mL tube and centrifuge at 600 x g for 5 min.



2. Red blood cell lysis step
 - Remove serum free media and resuspend the cell pellet in **1 mL of red blood cell (RBC) lysis buffer for 3 min** (too much time in the RBC lysis buffer will hurt stem cells). RBC lysis buffer: sterile 0.83% ammonium chloride (room temperature) or bottle purchased from Millipore Sigma (#cat:11814389001, 4°C)
 - Cells are centrifuged at 600 x g for 5 min and RBC buffer is removed
 - Cells were resuspended in serum free media and counted

3. Plate bone marrow cells to each plate at 3 million cells per plate with 10 mL of BMDM differentiation media

- **L929 differentiation media (regular)** – RPMI 1640 contains 10% L929 (week1+week2 combined), 10% FBS, 2mM L-Glutamine (Thermo Fisher Scientific), 10 units and 100 $\mu\text{g mL}^{-1}$ penicillin-streptomycin (Sigma-Aldrich), 20 mM HEPES (Thermo Fisher Scientific, #cat:15630080), 50 μM beta-mercaptoethanol (Sigma-Aldrich, make fresh dilution filtered, 3.5 μL to 997 μL RPMI, store in -20°C). Add L929 right before plating.

***L929 is the preferred method because of cost.**

- **L929 differentiation media (SILAC)** -SILAC RPMI 1640 contains 10% L929 (week1+week2 combined), 10% DFBS, 2mM L-Glutamine (Thermo Fisher Scientific), 10 units and 100 $\mu\text{g mL}^{-1}$ penicillin-streptomycin (Sigma-Aldrich), 100 $\mu\text{g/mL}$ L-arginine, 100 $\mu\text{g/mL}$ L-lysine, 20 mM HEPES (Thermo Fisher Scientific, #cat:15630080), 50 μM beta-mercaptoethanol (Sigma-Aldrich, make fresh dilution filtered, 3.5 μL to 997 μL RPMI, store in -20°C). Add L929 right before plating.
- **MCSF differentiation media-** 20 ng/mL MCSF (Peprotech 315-02), 10% FBS, 2 mM L-Glutamine (Thermo Fisher Scientific), 10 units and 100 $\mu\text{g mL}^{-1}$ penicillin-streptomycin (Sigma-Aldrich), 50 μM beta-mercaptoethanol (Sigma-Aldrich), 20mM HEPES (pH 7.4)

4. On day 3, add an additional 10 mL of BMDM differentiation media.

5. On day 7, macrophage differentiation is completed.

L929 media collection

Corning™ CELLSTACK™ Culture Chambers - Tissue Culture Surface; 2 Chambers;
Total Culture Area: 1,272 cm^2 ; 5/Cs (Thermo Fisher 05-539-095)

Surface area calculated based on the number of cells seeded from leitingner lab.

1. L929 cells are cultured in complete RPMI (RPMI, 10% FBS, 2 mM L-Glutamine (Thermo Fisher Scientific), 10 units and 100 $\mu\text{g mL}^{-1}$ penicillin-streptomycin (Sigma-Aldrich), 20 mM HEPES (pH 7.4)).

Heat inactivation of FBS: 55 $^{\circ}\text{C}$ for 1hr (optional)

2. 678,400 cells will be seeded across 1 cell stack (2 chambers). Media amount 423 mL. Make sure the cells are distributed across both chambers evenly.
3. Allow the cells to grow at 37 °C for a week.
4. After a week, harvest the 1st week L929 media and replace the media with another 423mL of the RPMI complete media (RPMI, 10% FBS, 2mM L-Glutamine (Thermo Fisher Scientific), 10 units and 100µg mL⁻¹ penicillin-streptomycin (Sigma-Aldrich), 20mM HEPES (Thermo Fisher Scientific, #cat:15630080)).
5. After week 1, plate should be confluent with L929 cells. After another week of incubation, media is collected. Be careful not to disrupt the plate too much. Cells will detach from the plate.
6. After collecting second week of L929 media, filter the media through Stericup to remove any cell residue.
7. Aliquot and store L929 media in -80°C (thaw and refreeze at most once).

Revised on 08/01/2022